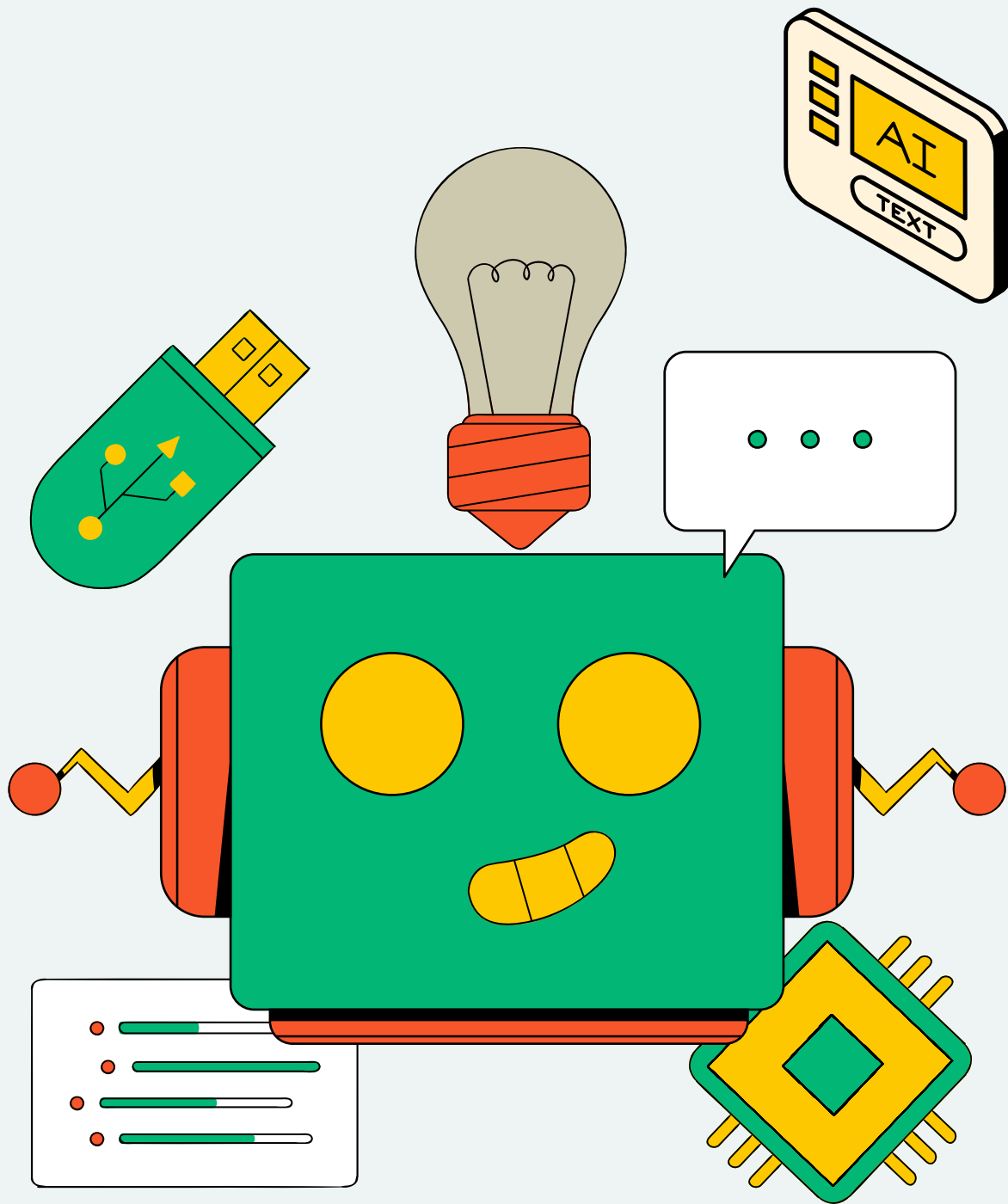
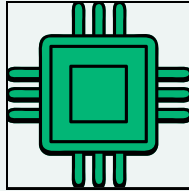


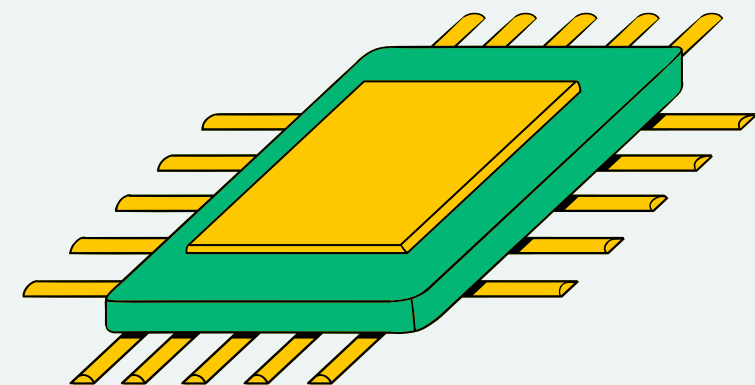


MACHINE LEARNING PROJECT - BANK CHURN PREDICTION

PRESENTATION

PRESENTED BY:

ALI ABOUHACHIM-ALAMI
MAXIME BENDAVID
YOUSSEF BENADDI



GOAL: IDENTIFY CUSTOMERS LIKELY TO CLOSE THEIR ACCOUNTS



DATASET OVERVIEW

RangeIndex: 165034 entries, 0 to 165033

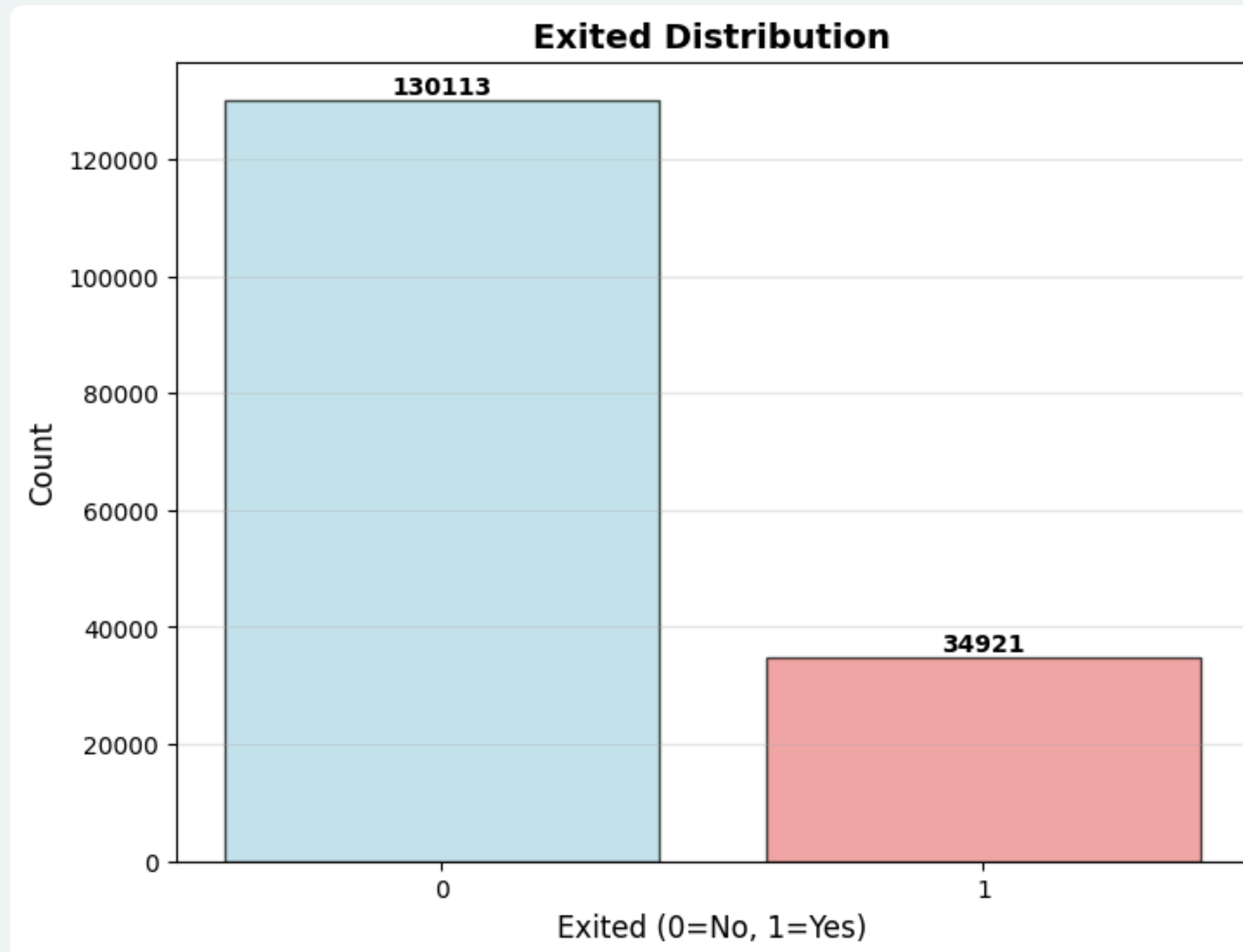
Data columns (total 14 columns):

#	Column	Non-Null Count	Dtype
0	id	165034 non-null	int64
1	CustomerId	165034 non-null	int64
2	Surname	165034 non-null	object
3	CreditScore	165034 non-null	int64
4	Geography	165034 non-null	object
5	Gender	165034 non-null	object
6	Age	165034 non-null	float64
7	Tenure	165034 non-null	int64
8	Balance	165034 non-null	float64
9	NumOfProducts	165034 non-null	int64
10	HasCrCard	165034 non-null	float64
11	IsActiveMember	165034 non-null	float64
12	EstimatedSalary	165034 non-null	float64
13	Exited	165034 non-null	int64

dtypes: float64(5), int64(6), object(3)



DATASET IMBALANCE

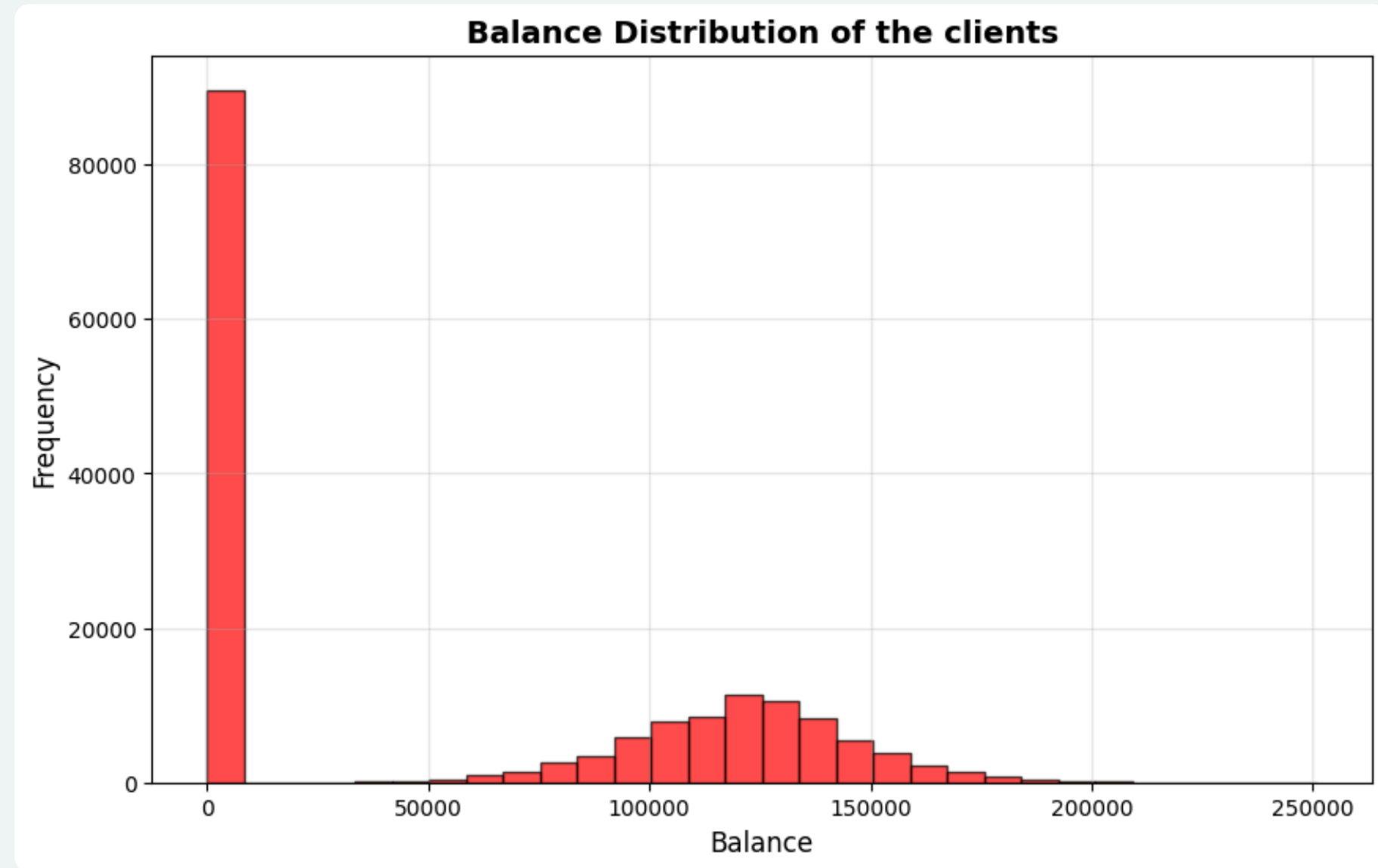


Solution:
SMOTE Algorithm



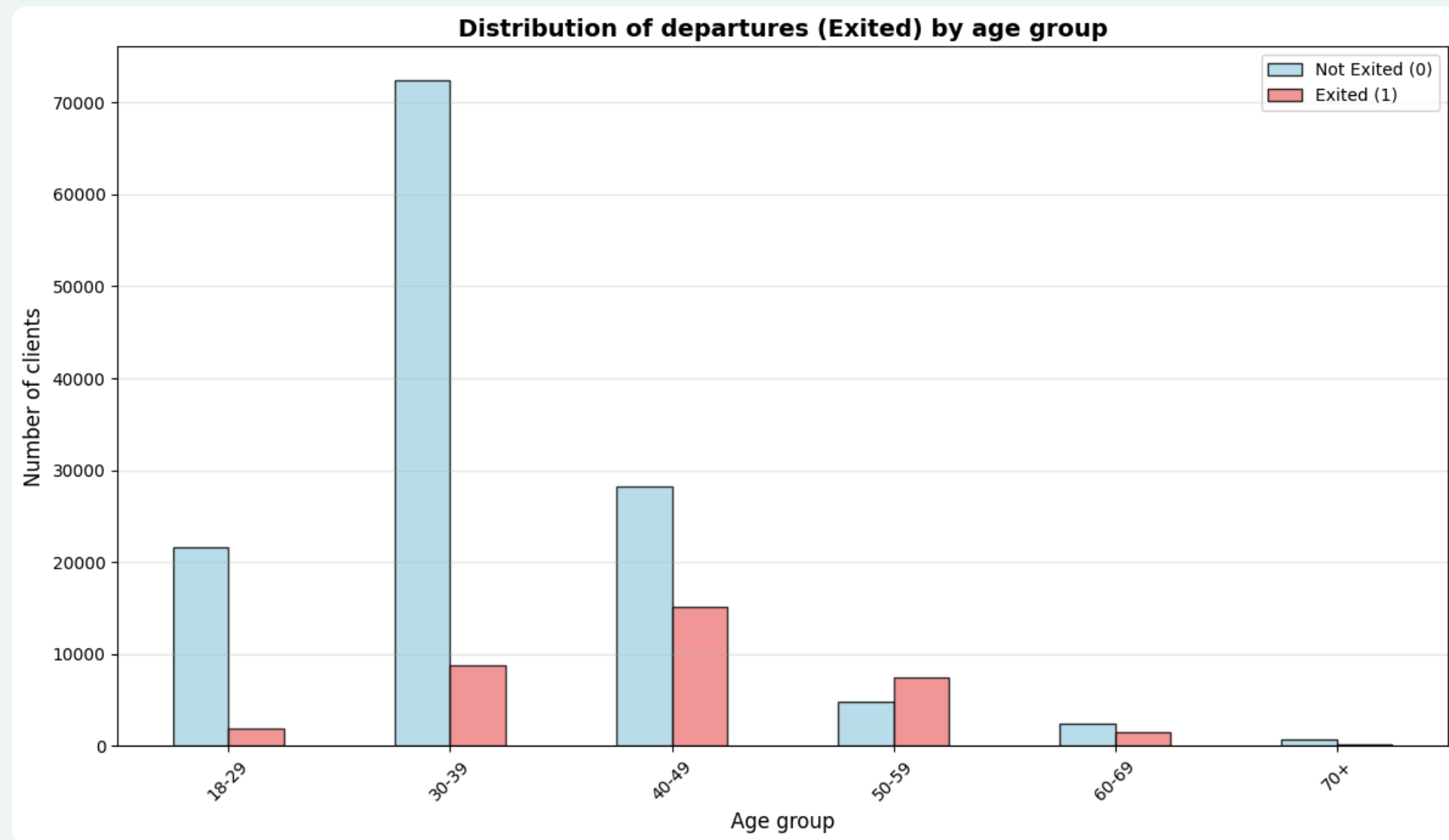
FEATURE ENGINEERING

- **IsZeroBalance**
- **TenurePerAge**
- **BalancePerSalary**
- **AgeGroup**



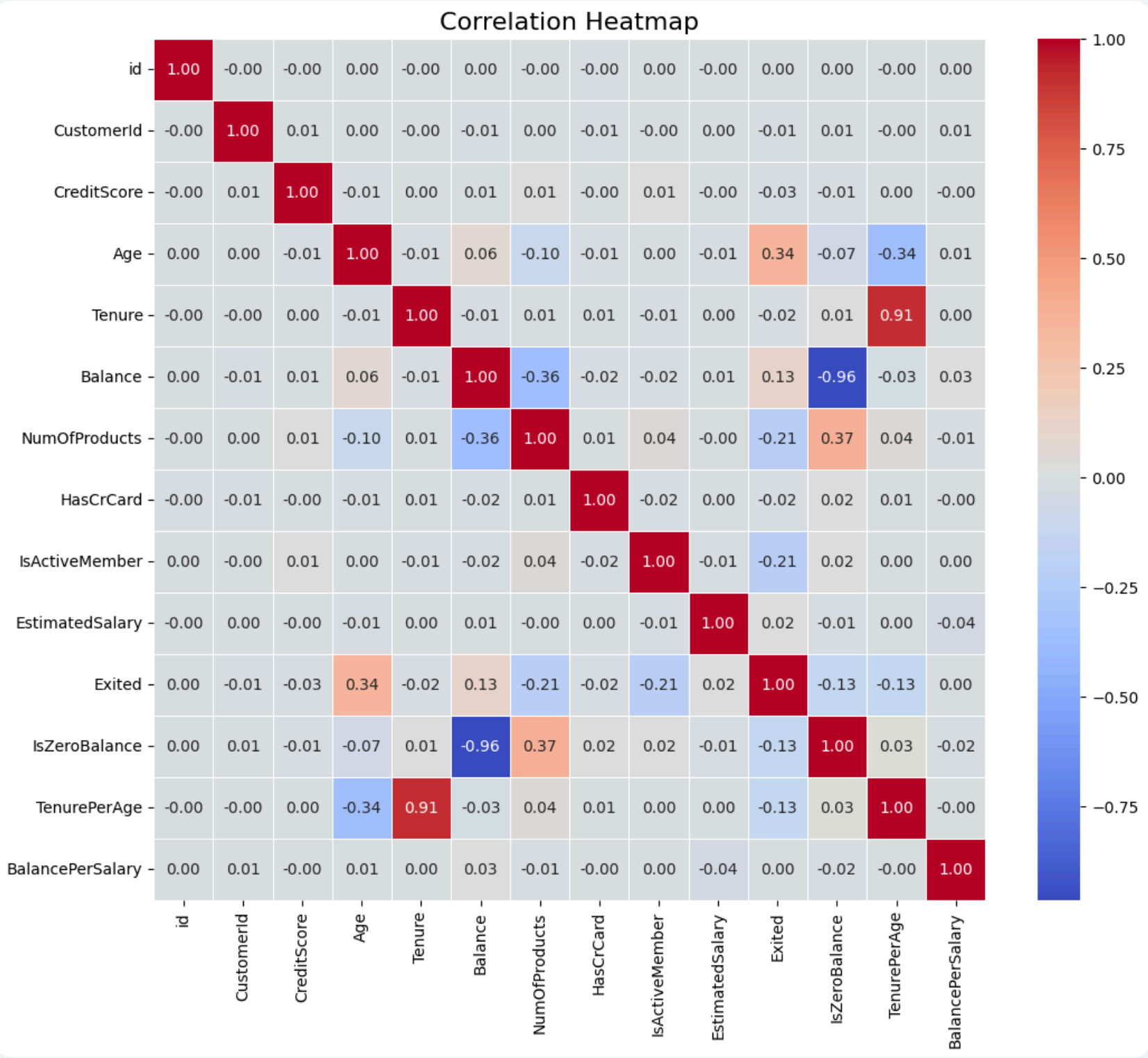
FEATURE ENGINEERING

- **IsZeroBalance**
- **TenurePerAge**
- **BalancePerSalary**
- **AgeGroup**



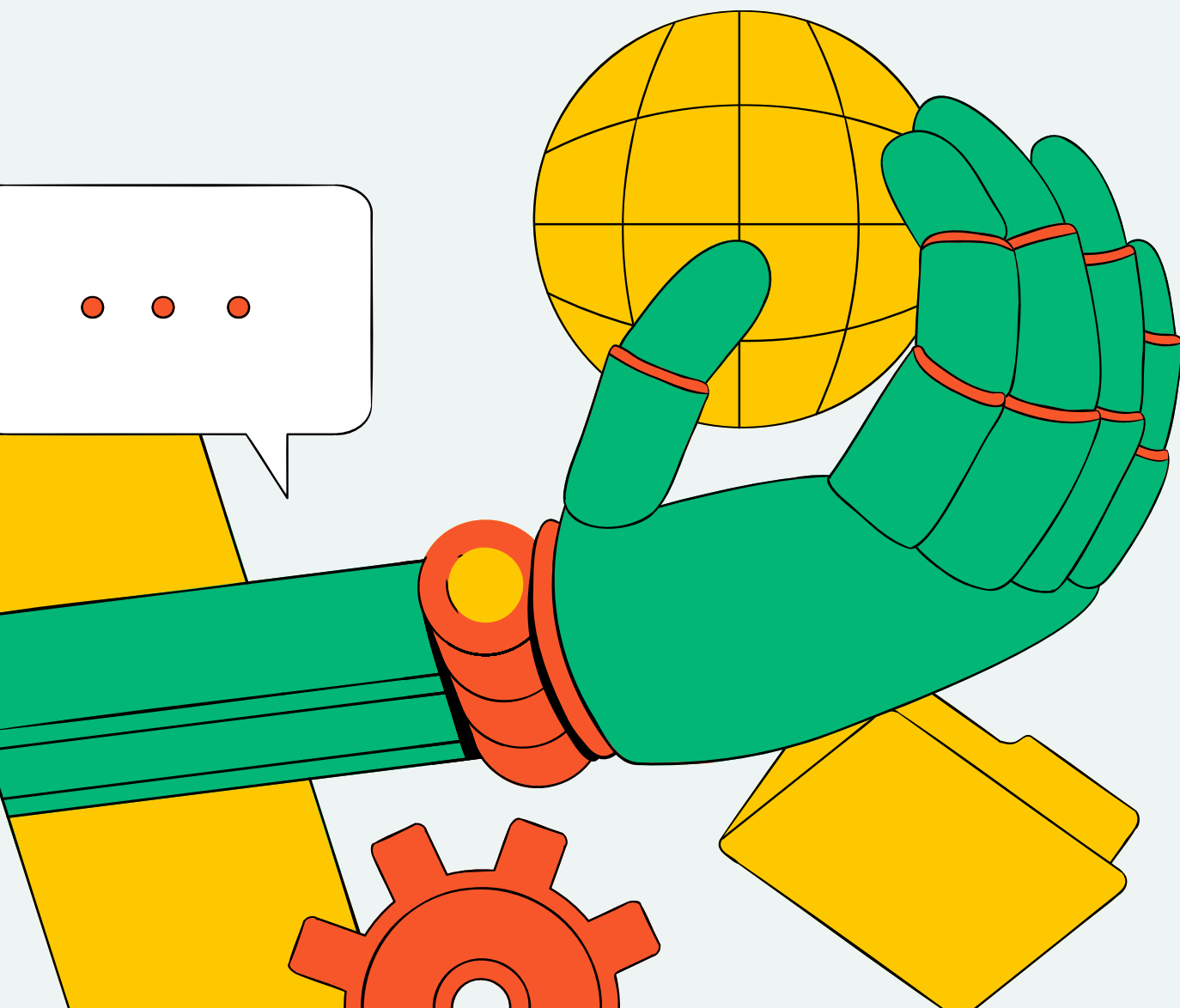
CORRELATION ANALYSIS

Exited	1.000000
Age	0.340768
NumOfProducts	0.214554
IsActiveMember	0.210237
IsZeroBalance	0.133346
Balance	0.129743
TenurePerAge	0.128395
CreditScore	0.027383
HasCrCard	0.022141
Tenure	0.019565
EstimatedSalary	0.018827
CustomerId	0.009947
id	0.002512
BalancePerSalary	0.001559
Name: Exited, dtype: float64	



METRIC FORMULA FOR F2-SCORE

$$F_{\beta} = (1 + \beta^2) \cdot \frac{\text{Precision} \cdot \text{Recall}}{(\beta^2 \cdot \text{Precision}) + \text{Recall}}$$



OUR MODELS

Logistic Regression, SVM

PCA

Tree Models (Decision Tree Classifiers)

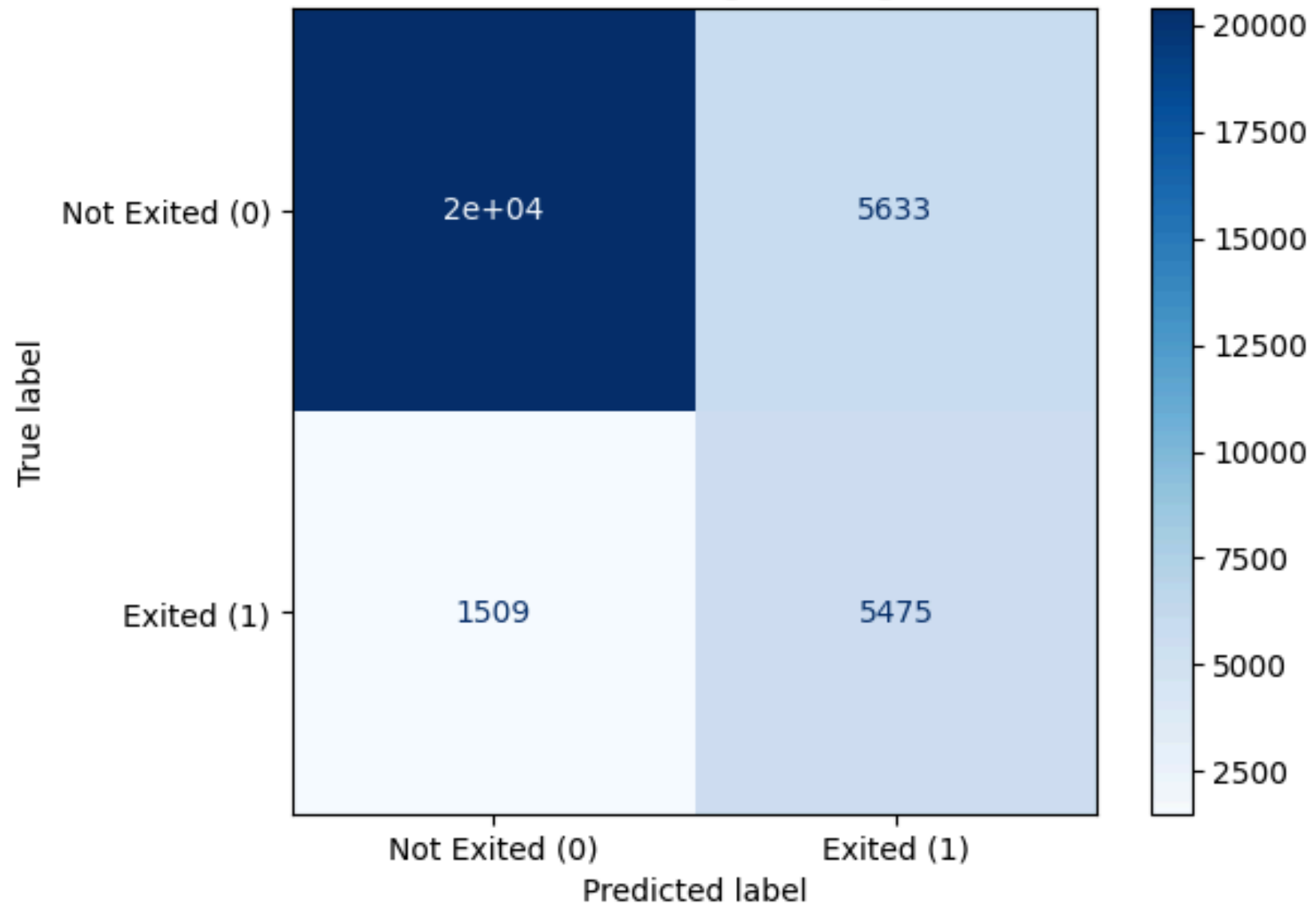
Ensemble models (Random Forest,
Gradient Boosting, Voting & Stacking
Classifier)

Neural Networks (MLP)

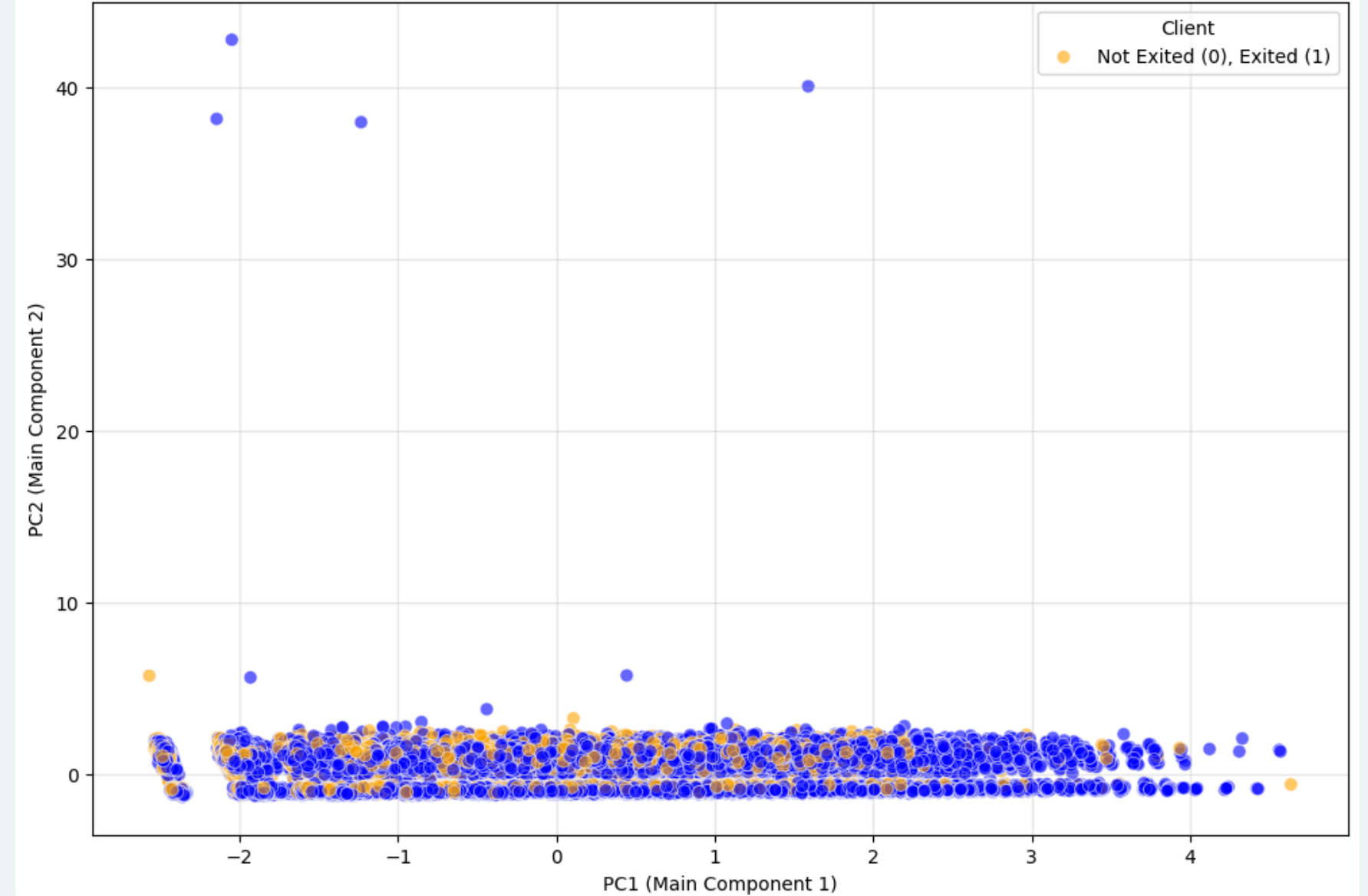


LOGISTIC REGRESSION - PCA

Confusion Matrix (Logistic Regression)

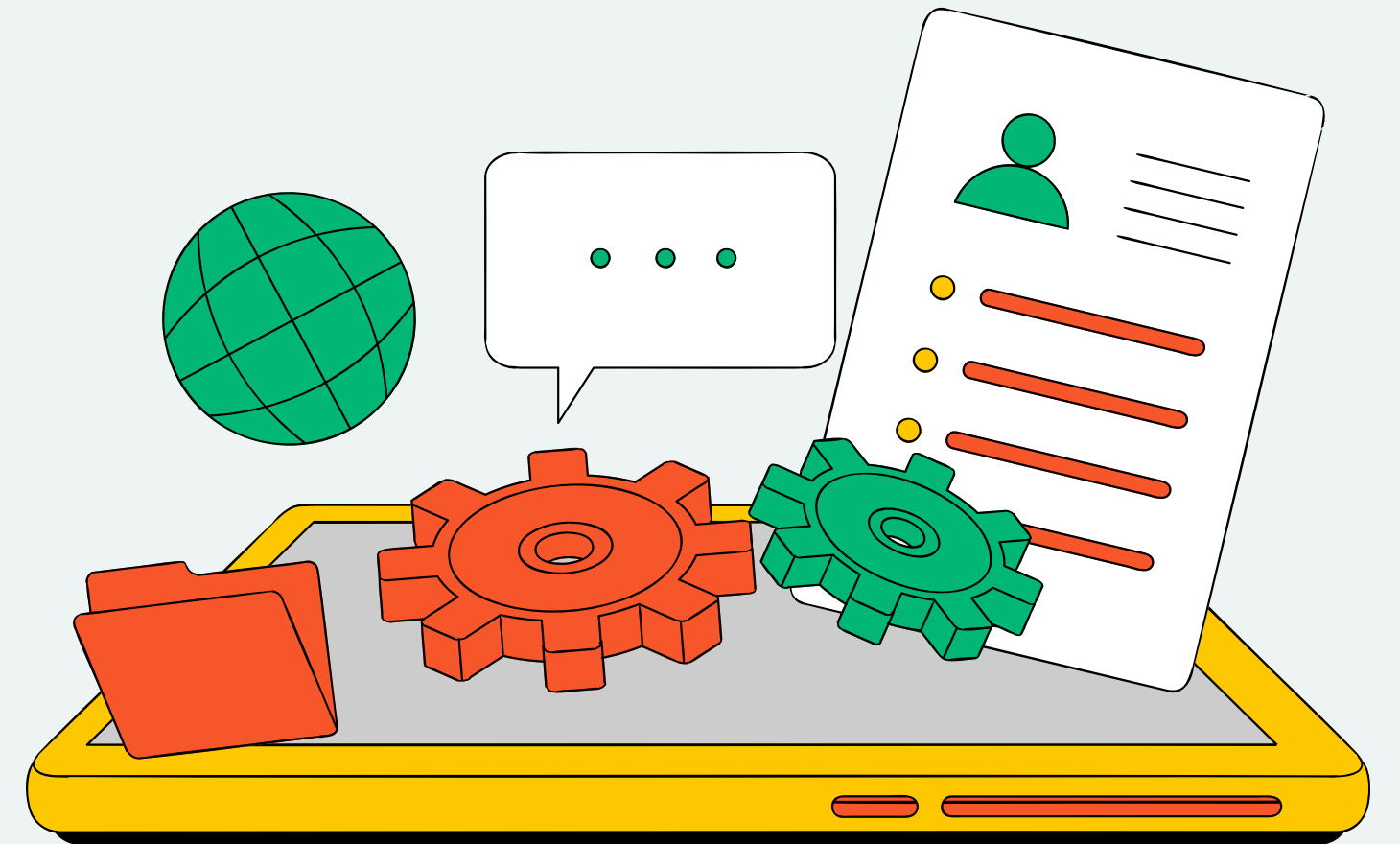


2D visualization of Customers (via PCA)



OBSTACLES

- **Execution Time**
- **Overfitting**
- **Data Imbalance**



OVERFITTING

```
fbeta Train : 0.9991  
fbeta Val   : 0.5406  
Gap        : 0.4585
```

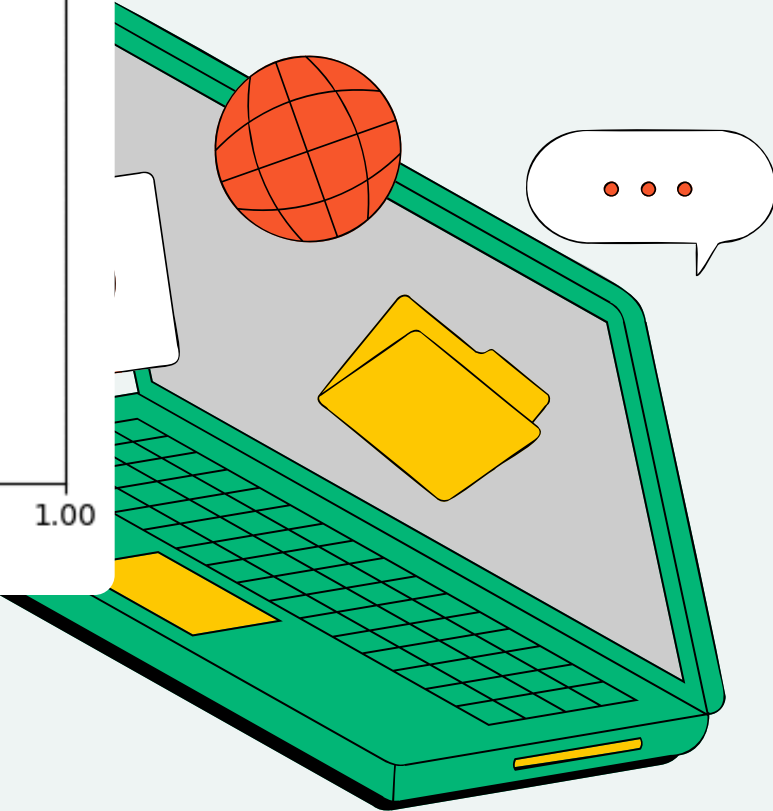
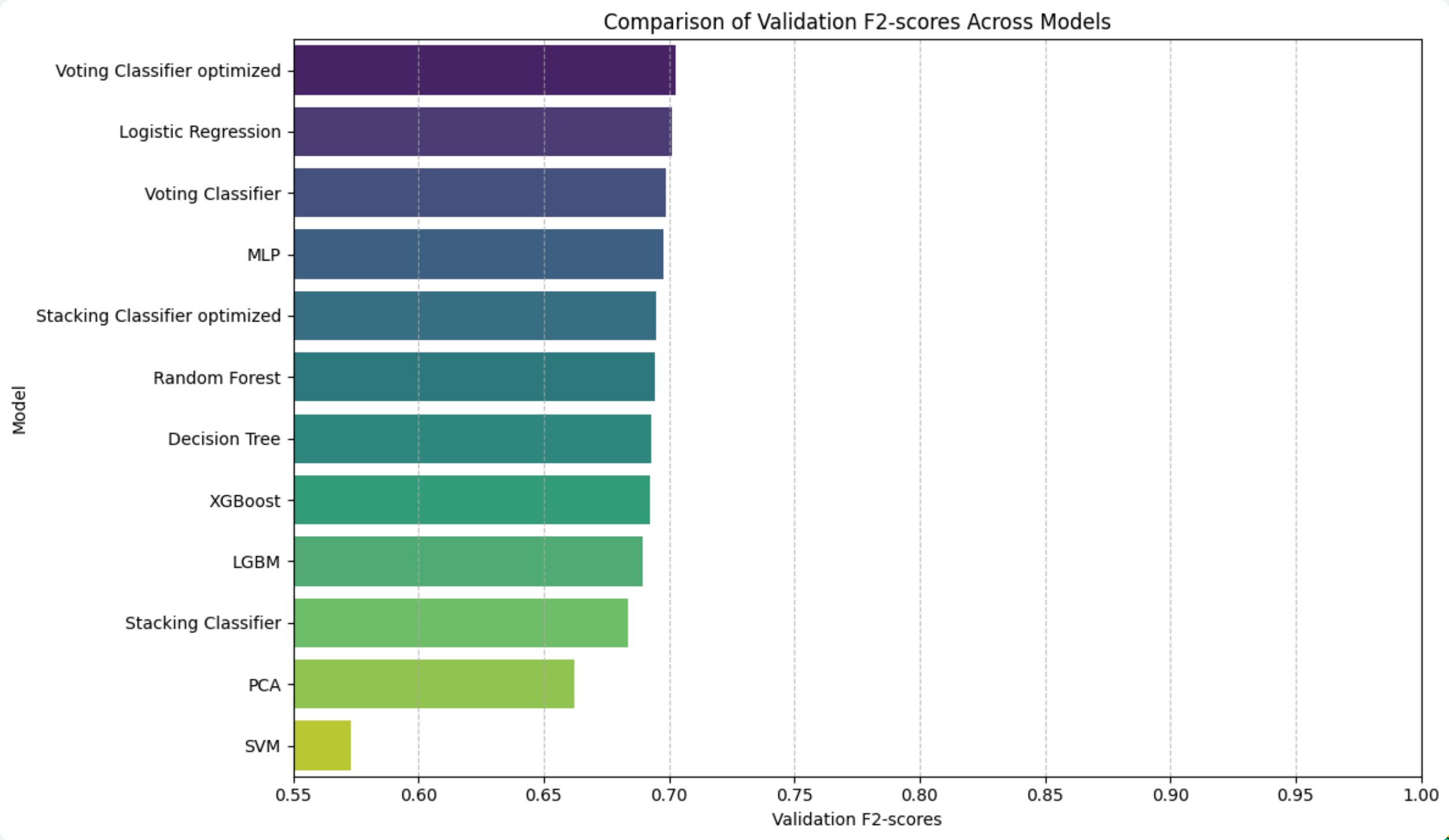
Solutions:

- Adjusting Hyperparameters
- Using Cross Validation: CV = 5
(then CV 3)



```
fbeta Train : 0.8145  
fbeta Val   : 0.6927  
Gap        : 0.1218
```

COMPARISON OF MODELS



FINAL TEST & CONCLUSION

F2-score of our best model on the test data: 0.698

